

## Course Syllabus

### General Information

|                   |                        |                      |           |
|-------------------|------------------------|----------------------|-----------|
| <b>Faculty</b>    | Information Technology |                      |           |
| <b>Department</b> | Cybersecurity Dept.    |                      |           |
| <b>Semester</b>   | 2 <sup>nd</sup>        | <b>Academic Year</b> | 2025/2026 |

### Course Details

|                                   |                                                                                                                                                     |             |   |              |   |         |  |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---|--------------|---|---------|--|
| Course Title                      | Cybersecurity Fundamentals                                                                                                                          |             |   | Course No.   |   | 0452301 |  |
| Type of Learning                  | <input checked="" type="checkbox"/> On- campus <input type="checkbox"/> Blended ( ..... Platform) <input type="checkbox"/> Online ( ..... Platform) |             |   |              |   |         |  |
| Credit Hours                      | 3                                                                                                                                                   | Theoretical | 3 | Practical    | 0 |         |  |
| Course Level (according to JNQF ) |                                                                                                                                                     |             | 6 | JNQF Hours*  |   | 12.15   |  |
| Pre-requisite                     | 0181503 – Programming fundamentals                                                                                                                  |             |   | Co-requisite |   |         |  |

### Course Instructor Information

|                                          |                     |                     |     |
|------------------------------------------|---------------------|---------------------|-----|
| <b>Instructor Name</b>                   | Dr. Hashim Jarrar   |                     |     |
| <b>Instructor E-mail</b>                 | a.hassan@meu.edu.jo |                     |     |
| <b>Course Time</b>                       |                     |                     |     |
| <b>Office Hours</b>                      |                     |                     |     |
| <b>Office Number</b>                     | N133                | <b>Office Phone</b> | 512 |
| <b>Lab Instructor Name –If assigned-</b> | --                  |                     |     |

### Sources and References

|                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------|
| <b>1. Textbook</b>                                                                                                     |
| Computer Security principles and Practices, William Stalling and Lawrie Brown, 2014 3 <sup>rd</sup> Edition , Pearson. |
| <b>2. References</b>                                                                                                   |
| Security+ Guide to Network Security Fundamentals, Mark Ciampa, 2018, ISBN 978-1-337-28878-1.                           |
| Cryptography and Network Security, William Stallings, 2016 7 <sup>th</sup> Edition, Pearson.                           |

### Teaching Methods

|             |              |            |              |
|-------------|--------------|------------|--------------|
| 1. Lectures | 2. Questions | 3. Reports | 4. Exercises |
|-------------|--------------|------------|--------------|

### Course Description

This course aims to provide comprehensive study to understand the fundamentals of cybersecurity. Topics include Introduction to cybersecurity: Assets, Vulnerabilities, Threats, Countermeasures, Security policy, Attacks, Security services (Confidentiality, Integrity, Availability, Non-repudiation, Accountability). Cryptography: symmetric-key, asymmetric cryptography, hash function, message authentication code, Digital signature, Digital certificates. User authentication: User authentication principles, Password-Based Authentication, Token-Based Authentication, Biometric Authentication, Security Issues for User Authentication. Access control: Access Control Principles, Subjects, Objects, and Access Rights, Discretionary Access Control, Example: UNIX File Access Control. Malicious Software: Types of Malicious Software (Malware), Advanced Persistent Threat, Viruses, Worms, Social Engineering, Spam, Trojans, Zombie, Bots. Denial-of-Service Attacks: DoS Attacks, Flooding Attacks, DDoS Attacks.

### Course Learning Outcomes (CLOs) and its correlation with Program Learning Outcomes (PLOs) and JNQF Descriptors

| No. | CLOs                                                                                       | PLOs | JNQF Descriptors |       |            |
|-----|--------------------------------------------------------------------------------------------|------|------------------|-------|------------|
|     |                                                                                            |      | Knowledge        | Skill | Competency |
| 1   | Identify the fundamental cybersecurity concepts.                                           | 1    | K1-I             |       |            |
| 2   | Define the main security services and countermeasures.                                     | 1    | K1-I             |       |            |
| 3   | Describe the main components and requirements of the cryptographic system.                 | 1    | P1-I             |       |            |
| 4   | Discuss the user authentication concepts and access control principles.                    | 1    | K1-I             |       |            |
| 5   | Identify the main types of Malicious Software (Malware).                                   | 1    | K1-I             |       |            |
| 6   | State the major technical security challenges and tools for computer and network security. | 2    |                  | S1-I  |            |

### Course Topics

| Week | Topics                                                                                                                                                                                                 | Lecture Type |                    | CLOs | Assessment                    |                       |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------------|------|-------------------------------|-----------------------|
|      |                                                                                                                                                                                                        | Lecture No.  | Lecture Type       |      | Assessment Tool               | Summative/ Formative  |
| 1    | <b>Ch1: Introduction to cybersecurity</b><br>Assets, Vulnerabilities, Threats, Countermeasures.                                                                                                        | 1, 2         | Lecture On- campus | 1    | Discussion & Questions        | Formative             |
| 2    | Security policy, Attacks, Security services (Confidentiality, Integrity, Availability, Non-repudiation, Accountability).                                                                               | 1, 2         | Lecture On- campus | 2    | Discussion & Questions        | Formative             |
| 3    | <b>Ch2: Cryptographic Tools</b><br>Classical Cryptography<br>Borut Force and frequency analysis attacks<br>Confidentiality with Symmetric Encryption<br>Key exchange problem<br>Key management problem | 1, 2         | Lecture On- campus | 3    | Discussion & Questions        | Formative             |
| 4    | Integrity and Authenticity with Symmetric Encryption<br>Message Authentication Code and Hash Functions.                                                                                                | 1, 2         | Lecture On- campus | 3    | Discussion & Questions & Quiz | Formative & Summative |
| 5    | Public-Key Encryption<br>Digital Signatures and Key Management<br>Digital Signatures with hash functions<br>Digital Certificates                                                                       | 1, 2         | Lecture On- campus | 3    | Discussion & Questions        | Formative             |

|    |                                                                                                                                                                                  |      |                       |         |                                           |                             |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------------------|---------|-------------------------------------------|-----------------------------|
|    | Comparisons between Symmetric and Asymmetric Encryption                                                                                                                          |      |                       |         |                                           |                             |
| 6  | <b>Ch3: User authentication principles</b><br>Password-Based Authentication,<br>Token-Based Authentication,                                                                      | 1, 2 | Lecture<br>On- campus | 4       | Discussion<br>&<br>Questions              | Formative                   |
| 7  | Biometric Authentication,<br>Security Issues for User Authentication.                                                                                                            | 1, 2 | Lecture<br>On- campus | 4       | Discussion<br>&<br>Questions              | Formative                   |
| 8  | <b>Mid-Exam</b>                                                                                                                                                                  | 1, 2 | Assessment            | 1, 2, 3 | Questions<br>&<br>Rubric                  | Summative                   |
| 9  | <b>Ch4: Access control</b><br>Access Control Principles,<br>Subjects, Objects, and Access Rights                                                                                 | 1, 2 | Lecture<br>On- campus | 4       | Discussion<br>&<br>Questions              | Formative                   |
| 10 | Discretionary Access Control,<br>Example: UNIX File Access Control.                                                                                                              | 1, 2 | Lecture<br>On- campus | 4       | Discussion<br>&<br>Questions<br>&<br>Quiz | Formative<br>&<br>Summative |
| 11 | <b>Ch5: Malicious Software</b><br>Types of Malicious Software (Malware),<br>Advanced Persistent Threat , Viruses , Worms ,                                                       | 1, 2 | Lecture<br>On- campus | 5       | Discussion<br>&<br>Questions              | Formative                   |
| 12 | Social Engineering attack,<br>Spam , Trojans, Zombie, Bots.                                                                                                                      | 1, 2 | Lecture<br>On- campus | 5       | Discussion<br>&<br>Questions              | Formative                   |
| 13 | <b>Ch6 Denial-of-Service Attacks</b><br>DoS Attacks, Flooding Attacks, DDoS Attacks.                                                                                             | 1, 2 | Lecture<br>On- campus | 5       | Discussion<br>&<br>Questions              | Formative                   |
| 14 | <b>Ch7: Firewalls and Intrusion Prevention Systems</b><br>The Need for Firewalls,<br>Firewall Characteristics and Access Policy.                                                 | 1, 2 | Lecture<br>On- campus | 6       | Discussion<br>&<br>Questions              | Formative                   |
| 15 | Intrusion Detection: Intruders, Intrusion Detection,<br>Analysis Approaches. Web security: SSL and TLS,<br>Electronic mail security.<br><br><b>Research Project Presentation</b> | 1, 2 | Lecture<br>On- campus | 6       | Discussion<br>&<br>Questions              | Formative                   |
|    |                                                                                                                                                                                  |      | Lecture<br>On- campus | 1       | Questions<br>&<br>Rubric                  | Summative                   |
| 16 | <b>Final Exam</b>                                                                                                                                                                | -    | Assessment            | 1..6    | <b>Questions<br/>&amp;<br/>Rubric</b>     | <b>Summative</b>            |

| Assessment Tool       | Weight % | Expected Due Date |
|-----------------------|----------|-------------------|
| Mid Exam              | 25%      | Week 8            |
| Quizzes               | 10%      | Week 4, 10        |
| Assignments           | 10%      | Week 7, 12        |
| Research/Presentation | 15%      | Week 15           |
| Final Exam            | 40%      | Week 16           |

### Course Policies

| Item                        | Policy                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Quizzes</b>              | <ul style="list-style-type: none"> <li>No makeup quizzes.</li> <li>A minimum 2 quizzes are given.</li> <li>Each Quiz is out of 5.</li> <li>If five quizzes or more are given then the lowest quiz's grade is dropped.</li> </ul>                                                                                                                                                    |
| <b>Exams</b>                | <ul style="list-style-type: none"> <li>The format of the exams is generally (but NOT always) as follows: General Definitions, Multiple-Choice, True/False, Analyze a Problem, Essay Questions, etc.</li> </ul>                                                                                                                                                                      |
| <b>Makeup Exams</b>         | <ul style="list-style-type: none"> <li>Makeup exam should not be given unless there is a valid excuse.</li> </ul>                                                                                                                                                                                                                                                                   |
| <b>Drop Date</b>            | <ul style="list-style-type: none"> <li>Last day to drop the course is according to the University regulations.</li> </ul>                                                                                                                                                                                                                                                           |
| <b>Academic Honesty</b>     | <ul style="list-style-type: none"> <li>Cheating or copying on exam or quiz is an illegal and unethical activity.</li> <li>Standard MEU policy will be applied.</li> <li>All graded assignments must be your own work (your own words).</li> </ul>                                                                                                                                   |
| <b>Attendance</b>           | <ul style="list-style-type: none"> <li>Excellent attendance is expected.</li> <li>MEU policy requires the faculty member to assign ZERO grade if a student misses 20% of the classes.</li> <li>Sign-in sheets will be circulated.</li> <li>If you miss class, it is your responsibility to find out about any announcements or assignments you may have missed.</li> </ul>          |
| <b>E-Learning</b>           | <ul style="list-style-type: none"> <li>It is your responsibility to check the course's eLearning website regularly (at least once every 2-3 days) for announcements and material.</li> </ul>                                                                                                                                                                                        |
| <b>Workload</b>             | <ul style="list-style-type: none"> <li>The student should expect to spend 6 hours per week as an average workload.</li> </ul>                                                                                                                                                                                                                                                       |
| <b>Graded Exams</b>         | <ul style="list-style-type: none"> <li>Instructor should return exam papers graded to students within one week after the exam date.</li> </ul>                                                                                                                                                                                                                                      |
| <b>Participation</b>        | <ul style="list-style-type: none"> <li>Participation in and contribution to class discussions will affect your final grade positively. Raise your hand if you have any questions.</li> <li>Making any kind of disruption and (side talks) in the class will affect you negatively.</li> </ul>                                                                                       |
| <b>Assignment/ Projects</b> | <ul style="list-style-type: none"> <li>Assignments are as mentioned in the previous table.</li> <li>Students organize themselves in teams of (2-3) students each. Each team will give at least one presentation regarding a selected case, delivering a fully working application.</li> <li>More details about phases and topics of the projects will be given later on.</li> </ul> |
| <b>Others</b>               |                                                                                                                                                                                                                                                                                                                                                                                     |

**Rubrics and Feedbacks (If Applicable)**

**\* JNQF Hours Calculation**

| Activity                                        | Duration/ week          | Times/ Semester | No. of Hours |
|-------------------------------------------------|-------------------------|-----------------|--------------|
| <b>Learning Activities</b>                      |                         |                 |              |
| Lectures and Seminars                           | 3 hours                 | 15              | 45           |
| Tutorials                                       | 0                       | 0               | 0            |
| laboratory                                      | 0                       | 0               | 0            |
| <b>Activities for Assessment and Evaluation</b> |                         |                 |              |
| First Exam/ Midterm                             | 1.5 hours               | 1               | 1.5          |
| Second Exam (If applicable)                     | 0                       | 0               | 0            |
| Final Exam                                      | 2 hours                 | 1               | 2            |
| Quizzes                                         | 15 minutes              | 2               | 0.5          |
| <b>Self-Learning Activities</b>                 |                         |                 |              |
| Assignments/ Homework                           | 2.5 hours               | 2               | 5            |
| Case Studies                                    | 0                       | 0               | 0            |
| Presentations                                   | 0                       | 0               | 0            |
| E-learning                                      | 0                       | 0               | 0            |
| Extra Readings                                  | 4.5                     | 15              | 67.5         |
| <b>Total Hours</b>                              | <b>121.5</b>            |                 |              |
| <b>JNQF Hours</b>                               | <b>=121.5/10 =12.15</b> |                 |              |